

ISO465 - Gitonga, Lee Njiru

Circuit design

1. Calculate the total resistance in ohms

a) In series

$$= R_1 + R_2 + R_3$$

$$\begin{aligned} & (20 + 30 + 50) \Omega \\ & = (20 + 80) \Omega \\ & = 100 \Omega \end{aligned}$$

b) In parallel

$$\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{0.08}$$

$$= 12.5 \Omega$$

$$\frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$\begin{aligned} & = 0.05 + 0.01 + 0.02 (\Omega) \\ & = (0.06 + 0.02) \Omega \\ & = 0.08 \Omega \end{aligned}$$

2. Calculate the total current from the power supply

$$\begin{aligned} \text{a) Series} \quad V &= I \times R & \frac{100 \Omega}{0.08 \Omega} \\ V/R &= I & 125 \text{ V} / 100 \Omega \\ & & = 1.25 \text{ A} \end{aligned}$$

$$b) \frac{V}{R} = I : \frac{125V}{0.08\Omega} = 1562.5A$$

$$\frac{125V}{12.5\Omega} = 10A$$

3. Calculate the current through each sensor

a) Series

$$\frac{125}{20} = 6.25A$$

$$\frac{125}{30} = 4.167A$$

$$\frac{125}{50} = 2.5A$$

1.25A for all (I)

b) Parallel

$$\frac{125}{20\Omega} = 6.25A (I_1)$$

$$\frac{125}{100\Omega} = 1.25A (I_2)$$

$$\frac{125}{50\Omega} = 2.5A (I_3)$$

$$\text{Confirm} = 6.25A + 1.25A + 2.5A$$

$$= 10A$$

4. Calc voltage drop through each resistor

a) $I \quad R \quad V$
 Series $1.25 \times 20 = 25$
 $1.25 \times 30 = 37.5$
 $1.25 \times 50 = 62.5$

$$25 + 37.5 + 62.5 = 125$$

b) ~~125V~~
 Parallel ~~125A~~
 $R \quad R \quad I$
 $12.5 \times 10 = 125V$

5. Calculate the power dissipated through each resistor (Power = Voltage * Current)

a) $P = V \times I$
 Series

$$= 125 \times$$

~~$P_x =$~~

$$P = \frac{I}{A} \times \frac{V}{V} = 156.25 \text{ W}$$

~~P_1~~

(A) (I)

$$25 \times 1.25 = 31.25 \text{ W}$$

~~P_2~~

$$37.5 \times 1.25 = 46.875 \rightarrow 46.88 \text{ W}$$

$$62.5 \times 1.25 = 78.125 \rightarrow 78.13 \text{ W}$$

Parallel
b)

$$P_1 = 6.25 \text{ A} \times 125 = 781.25 \text{ W}$$

$$P_2 = 1.25 \text{ A} \times 125 = 156.25 \text{ W}$$

$$P_3 = 2.5 \text{ A} \times 125 = 312.5 \text{ W}$$

PART B

- ① Develop a schematic diagrams that describe the 2 circuits in Part A and include tables that list all components per schematic

Series a) Components

- a) Resistor
- b) Power supply
- c) Wiring

Parallel

b) Components

- a) Resistor
- b) Power supply
- c) Wiring